

Rohan Anil Shinde

Machine Learning Engineer | AI/ML Systems Developer

PUNE, INDIA
Immediate Availability
EU Blue Card / Visa Eligible

rohitshinde3903@gmail.com • +91 7499273903 • rohiit.is-a.dev • linkedin.com/in/rohit-shinde • github.com/rohitshinde3903

Profile | Kurzprofil

Machine Learning Engineer with 4+ years of hands-on experience designing, training, and deploying production AI systems across the complete software lifecycle. Strong foundation in statistical modeling, NLP, computer vision, and deep learning. Specialized in fine-tuning open-weights LLMs (QLoRA, PEFT), curriculum-grounded RAG pipelines, and containerized low-latency microservices with Python, FastAPI, Docker, and GCP.

Beruflicher Werdegang | Professional Experience

12/2025 – 06/2026	Machine Learning / AI Engineer EduAI Hub	Pune, India
	- Directed a domain-specific LLM system; fine-tuned Gemma 3B via LoRA, QLoRA and progressive layer unfreezing, improving Q&A domain accuracy by 38%. - Architected RAG pipelines with LlamaIndex and ChromaDB, reducing hallucinations by 42% against curriculum documents. - Reduced inference latency by 40% using parameter-efficient quantization and high-throughput vLLM serving. - Built MLOps workflows with MLflow and Weights & Biases; deployed FastAPI/Docker services on GCP Cloud Run through CI/CD.	
01/2024 – Present	Freelance ML / AI Engineer Independent Projects	Remote
	- Delivered end-to-end ML, NLP and intelligent automation systems for international clients from discovery through production deployment. - Engineered Python data-processing pipelines, reducing operational data-handling overhead by 70%. - Built production REST APIs and asynchronous inference endpoints using FastAPI and Celery.	
06/2021 – Present	Founder & Lead Engineer Stones Web Services	Pune, India
	- Founded a digital solutions agency delivering cloud architectures and ML integrations for 15+ startups and regional businesses. - Built production web apps and APIs with Python (FastAPI, Django) and React, integrating NLP and computer-vision modules. - Managed containerized GCP infrastructure, deployment workflows and automated CI/CD.	
11/2024 – 04/2025	Machine Learning Researcher / Developer Dr. D. Y. Patil SOST	Pune, India
	- Developed ML, computer-vision and anomaly-detection algorithms for an AI-assisted electronic voting and identity-verification system. - Investigated facial verification and diffusion-based synthetic data generation for fraud-detection stress testing.	
08/2024 – 11/2024	Technical Trainer (Python & Web Frameworks) Teknowell EduTech	Pune, India
	Instructed 50+ students in Python, Django, Flask, REST APIs, Git and deployment; supervised capstone labs.	

Ausgewählte Projekte | Selected Projects

2025 – 2026	Domain-Specific LLM with RAG Architecture EduAI Hub	Production
Stack: PyTorch • Hugging Face • LlamaIndex • ChromaDB • FastAPI • GCP • MLflow		
■ Fine-tuned Gemma 3B via QLoRA; reduced hallucinations by 42% through vector retrieval and reranking.		
■ Deployed scalable inference endpoints on GCP Cloud Run with latency logging and MLflow evaluation.		
2024 – 2025	AI-Powered Biometric Authentication Platform Research Initiative	Prototype
Stack: Python • OpenCV • FastAPI • Django • Scikit-learn • Docker		
■ Designed multi-factor verification with anti-spoofing facial recognition and real-time behavioral anomaly scoring.		
■ Implemented asynchronous microservices for low-latency responses under simulated concurrent loads.		
2025 – 2026	Automated Educational Diagram Generation Engine Stones Web Services	Production
Stack: PyTorch • Stable Diffusion XL • ControlNet • LoRA • FastAPI • Docker		
■ Built a controlled generative pipeline producing structured educational diagrams from text using ControlNet edge conditioning.		

Ausbildung | Education

08/2021 – 06/2025	Bachelor of Technology (B.Tech) in AI & Data Science Dr. D. Y. Patil Vidyapeeth	Pune, India
CGPA: 9.45 / 10.0 (German Equivalent: Note ≈ 1,1 via Modified Bavarian Formula)		

Core Modules:
Machine Learning,
Deep Neural
Networks,
Probability &
Applied Statistics,
Computer Vision,
Natural Language
Processing, Data
Structures &
Algorithms.

Fachliche Kompetenzen | Technical Competencies

Programming	Python (Core), SQL, TypeScript, JavaScript, C++, Bash
ML & Math	PyTorch, TensorFlow, Scikit-learn, Hugging Face, SciPy, NumPy, Pandas, Statistical Modeling
GenAI & LLMs	RAG, Fine-Tuning (LoRA, QLoRA, PEFT), vLLM, Prompt Engineering, LlamaIndex, LangChain
Vector Search	ChromaDB, FAISS, Milvus, Qdrant, Semantic Embeddings, Hierarchical Chunking
MLOps & Cloud	Docker, GCP (Vertex AI, Cloud Run, GCS), MLflow, W&B, Git, GitHub Actions (CI/CD)
Backend	FastAPI, Django, Flask, asyncio, REST APIs, Celery, Redis, PostgreSQL

Zertifikate | Certifications

Google	Google Advanced Data Analytics Professional Certificate (Statistical Modeling & Predictive ML)
Coursera	The Power of Statistics: Inference, Hypothesis Testing & Mathematical Modeling

Sprachkenntnisse & Status | Languages & Mobility

Languages	English (Full Professional Proficiency, C1), German (Elementary, A1 in progress), Hindi (Bilingual), Marathi (Native)
Work Authorization	Indian Citizen Immediate Availability Eligible for German EU Blue Card / Fast-Track Skilled Worker Visa